
Project name : thumb bell verification system

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1. Name

Thumb Bell Verification System

A Smart Home Door Bell & Fingerprint Verification System

2. Background

Traditional doorbells only notify the homeowner that someone is at the door. They do not verify who the visitor is. The Thumb Bell Verification System is designed for home use to provide an additional layer of security by using fingerprint/thumb verification. Registered family members can be recognized through their fingerprints, while unregistered fingerprints can trigger an alert and keep the door locked.

3. Objective

The main objectives of this system are to:

- Provide a secure and smart home door-bell system.
- Verify registered users through their fingerprints.
- Allow access only to authorized family members.
- Keep the door locked for unregistered users.
- Record successful and failed verification attempts.
- Provide an easy-to-use dashboard for managing users and security settings.

4. Stakeholders

- Homeowner/Admin: Manages users, security settings, and system records.
- Family Members: Register their fingerprints and use the system for authorized entry.
- Visitors: Use the doorbell but are not granted access unless authorized.
- System Developer: Develops, maintains, and updates the software.
- Hardware/System: Fingerprint sensor, door lock, and doorbell that work with the software.

5. Requirements

Functional Requirements

1. User/Admin login.
2. Register new family members.
3. Scan and store fingerprints.
4. Verify fingerprints when someone uses the doorbell.
5. Grant access to registered users.
6. Deny access to unregistered users.
7. Activate the doorbell.
8. Maintain verification history.
9. Display successful and failed attempts.
10. Allow Admin to edit/delete registered users.
11. Provide security and notification settings.

Non-Functional Requirements

- Security: Fingerprint data and user information should be protected.
- Usability: The interface should be simple and easy to understand.
- Reliability: The system should respond correctly to fingerprint verification.
- Performance: Verification should be completed quickly.

- **Availability:** The system should be available whenever the home security system is active.
- **Maintainability:** The software should be easy to update and maintain.

